

🚀 Kubernetes Lab Challenge: Multi-Container Pod (Sidecar Pattern)

🎯 Objective

Design and deploy a multi-container pod in your existing EKS cluster using the **Sidecar pattern**.

The pod should:

- Include a **main container** running your web application
- Include a **sidecar container** that performs an auxiliary function (like logging, proxying, or static file serving)

💡 What You'll Learn

- Running multiple containers in a single pod
- Understanding inter-container communication
- Applying Kubernetes design patterns

🧐 Challenge Description

You are asked to:

1. **Containerize a simple web app** in a language of your choice (Python, Node.js, Java, etc.)
2. Create a **sidecar container** in the same pod that either:
 - Logs traffic (e.g., using busybox)
 - Proxies traffic (e.g., using nginx)
 - Adds value to your main app (e.g., metrics scraper)
3. Deploy both containers **within the same pod**
4. Verify that both containers are running and interacting correctly

🛠️ Requirements

- Choose any stack you're comfortable with (Flask, Express, Spring Boot, etc.)
- Use Kubernetes manifests (YAML) to define your pod
- Make the pod accessible locally via `kubectl port-forward`
- Confirm both containers are running and functional

💬 Hints

- Use a base image like `python:3.9`, `node:18`, `openjdk`, etc.
- Refer to these resources:
 - <https://kubernetes.io/docs/concepts/workloads/pods/#using-multiple-containers>
 - <https://learnk8s.io/design-patterns#sidecar>

-  https://hub.docker.com/_/nginx
- Define your containers under the same `spec.containers` block in the pod manifest
- To debug or inspect logs, use:
``bash
kubectl logs <pod-name> -c <container-name>
``

Completion Criteria

- You've deployed a pod with **two containers**
- You can access your web app
- Your sidecar container performs its job (e.g., logs or proxies traffic)

Bonus Tasks (Optional)

- Add a third container for health checks or metrics
- Use shared volumes to exchange files between containers
- Replace static ports with environment variables

Submission Instructions

Be ready to:

- Share your YAML manifest
- Show a working demo using `kubectl get pods`, `kubectl describe pod`, and `kubectl logs`